Shopping Cart System - Lab Work

1. Create a database called `ShoppingCartDB`.

```
CREATE DATABASE ShoppingCartDB;
```

2. Create a table titled `Customers` with the following constraints:

```
CREATE TABLE Customers (
 customer_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
 name VARCHAR(100) NOT NULL,
 email VARCHAR(100) UNIQUE NOT NULL,
 phone VARCHAR(20) NOT NULL,
 address TEXT NOT NULL
);
```

3. Create a table titled `Products` with the following fields:

```
CREATE TABLE Products (
 product_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
 product_name VARCHAR(100) NOT NULL,
 price DECIMAL(10,2) NOT NULL,
 stock INT NOT NULL,
 category VARCHAR(50) NOT NULL
);
```

4. Create a table titled `Orders`:

```
CREATE TABLE Orders (
 order_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
 customer_id INT NOT NULL,
 order_date DATE NOT NULL,
 total_price DECIMAL(10,2) NOT NULL,
 FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
);
```

5. Create a table titled `OrderItems`:

```
CREATE TABLE OrderItems (
 item_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
 order_id INT NOT NULL,
 product_id INT NOT NULL,
 quantity INT NOT NULL,
 FOREIGN KEY (order_id) REFERENCES Orders(order_id),
 FOREIGN KEY (product_id) REFERENCES Products(product_id)
);
```

Queries

Query 1 (Easy): Show all products under the "Electronics" category.

SELECT * FROM Products WHERE category = 'Electronics';

product_id	product_name	price	stock	category
1	Smartphone X	29999.99	10	Electronics
2	Bluetooth Speaker	2499.50	25	Electronics
NULL	NULL	NULL	NULL	NULL

Query 2 (Easy): List all orders made before June 15, 2025.

SELECT * FROM Orders WHERE order_date < '2025-06-15';

order_id	customer_id	order_date	total_price
1	1	2025-06-10	29999.99
2	2	2025-06-11	4798.50
3	3	2025-06-12	4398.00
4	4	2025-06-14	899.99
NULL	NULL	NULL	NULL

Query 3 (Medium): Show name and email of customers who made any orders.

SELECT DISTINCT C.name, C.email
FROM Customers C
JOIN Orders O ON C.customer_id = O.customer_id;

name	email
Alice Smith	alice@gmail.com
John Doe	john.doe@example.com
Rina Akter	rina@outlook.com
Mizan Rahman	mizan@mail.com
Sadia Islam	sadia@yahoo.com

Query 4 (Medium): Find the product names and quantities for order ID = 1.

SELECT P.product_name, OI.quantity
FROM OrderItems OI
JOIN Products P ON OI.product_id = P.product_id
WHERE OI.order_id = 1;

product_name	quantity
Smartphone X	1

Query 5 (Medium-Hard): Find the total amount spent by each customer.

```
SELECT C.name, SUM(O.total_price) AS total_spent
FROM Customers C
JOIN Orders O ON C.customer_id = O.customer_id
GROUP BY C.customer_id;
```

name	total_spent
Alice Smith	29999.99
John Doe	4798.50
Rina Akter	4398.00
Mizan Rahman	899.99
Sadia Islam	1299.00

Query 6 (Medium-Hard): Find the most recently placed order with customer name.

```
SELECT C.name, O.order_date
FROM Orders O
JOIN Customers C ON O.customer_id = C.customer_id
ORDER BY O.order_date DESC
LIMIT 1;
```

name	order_date
Sadia Islam	2025-06-15

Query 7 (Medium-Hard): List products that have been ordered more than once.

```
SELECT P.product_name, COUNT(*) AS times_ordered
FROM OrderItems OI
JOIN Products P ON OI.product_id = P.product_id
GROUP BY P.product_id
HAVING COUNT(*) > 1;
```

product_name	times_ordered
Running Shoes	2

Sample Data Insertions

Customers Table

```
INSERT INTO Customers (name, email, phone, address) VALUES
('Alice Smith', 'alice@gmail.com', '+8801711000001', 'Dhaka, Bangladesh'),
('John Doe', 'john.doe@example.com', '+8801711000002', 'Rajshahi, Bangladesh'),
('Rina Akter', 'rina@outlook.com', '+8801711000003', 'Chittagong, Bangladesh'),
('Mizan Rahman', 'mizan@mail.com', '+8801711000004', 'Sylhet, Bangladesh'),
('Sadia Islam', 'sadia@yahoo.com', '+8801711000005', 'Khulna, Bangladesh');
```

Products Table

```
INSERT INTO Products (product_name, price, stock, category) VALUES
('Smartphone X', 29999.99, 10, 'Electronics'),
('Bluetooth Speaker', 2499.50, 25, 'Electronics'),
('Running Shoes', 3499.00, 15, 'Footwear'),
('Leather Wallet', 899.99, 30, 'Accessories'),
('Backpack', 1299.00, 20, 'Bags');
```

Orders Table

```
INSERT INTO Orders (customer_id, order_date, total_price) VALUES
(1, '2025-06-10', 29999.99),
(2, '2025-06-11', 4798.50),
(3, '2025-06-12', 4398.00),
(4, '2025-06-14', 899.99),
(5, '2025-06-15', 1299.00);
```

OrderItems Table

```
INSERT INTO OrderItems (order_id, product_id, quantity) VALUES
(1, 1, 1),
(2, 2, 1),
(2, 3, 1),
(3, 3, 1),
(3, 4, 1),
(5, 5, 1);
```